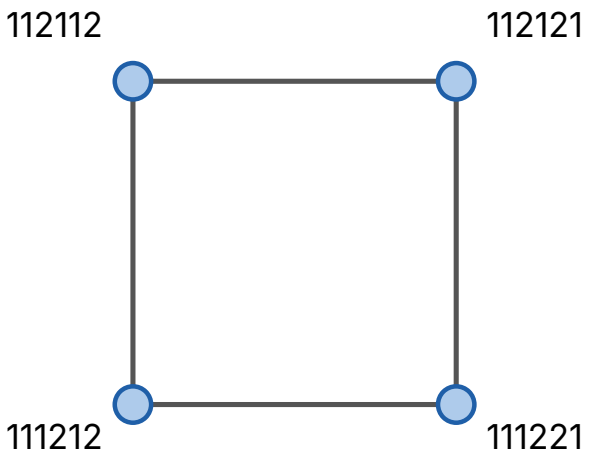
the 2-cube of histories $1 \to 72^2$ (Paper II Thm. cube)



all histories $1 \to 5184$: 16

combinatorial cube (4-cycle of legal swaps, filled as a 2-cell in the route complex)

homology of the full route complex needs more: loops and voids are recorded by Paper II (Thms. sphere, loopfam); not derived here